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Accent-Based Evaluation of Speech Anti-spoofing Countermeasures Across Multiple Languages

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Universidad de los Andes

THE HOOK · SAY IT, THEN PAUSE

Your mother calls. She needs $1,500, now. It is her voice.

— and it is not her.

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- **AI Lead Engineer**
  Konecto — voice agents, RAG and LLM systems in production
- **Researcher**
  FLAG Lab, Universidad de los Andes — speech anti-spoofing
- **Teaching assistant**
  Natural Language Processing, Universidad de los Andes

This talk is the first author's work. I present it, and I build on it.

Four moves, one conclusion

01 The attack

What a voice deepfake actually is, and the two ways to build one

02 The collapse

English-trained detectors meet Spanish speech

03 The culprit

Decomposing the failure by attack family

04 What survives

Bilingual training, and what is left after it

01

The attack

text-to-speech · voice conversion · why one of them is the dangerous one

Two ways to fake a voice

TTS

Text-to-speech

Writes the waveform from text. The speaker never said it — the vocoder invented it.

VC

Voice conversion

Takes a real recording and rewrites who is speaking. The words are real; the identity is not.

?

Why it matters

They leave different fingerprints — and only one of them survives a change of language.

One line of takeaway lives here. The supporting half of the sentence goes in grey.

Same model, different language

TRAINED ON ENGLISH, TESTED ON ENGLISH

2.67 %

EER · best configuration

Works. This is a solved problem — in English.

TRAINED ON ENGLISH, TESTED ON SPANISH

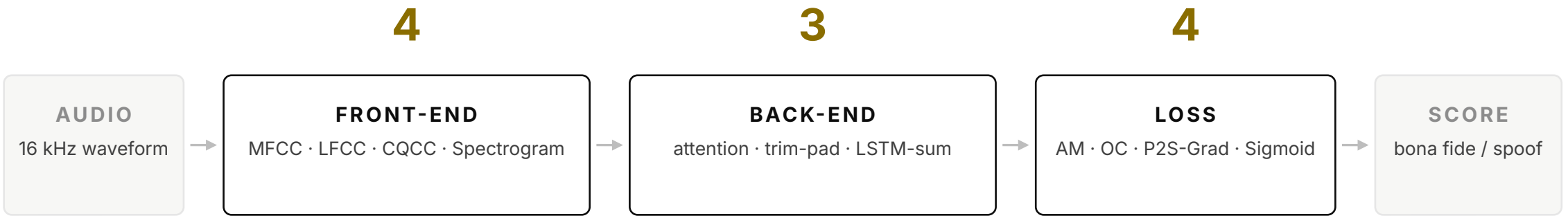
> 30 %

EER · all 48 configurations

Chance. Every single architecture we trained.

Tamayo et al. (2023) reported this. We went looking for why.

48 detectors, one pipeline



`4 × 3 × 4 = 48 architectures × 3 data conditions × 6 seeds = 864 trained models`

Every box that varies is counted. Nothing here is a single lucky run.

It is voice conversion. Not TTS.

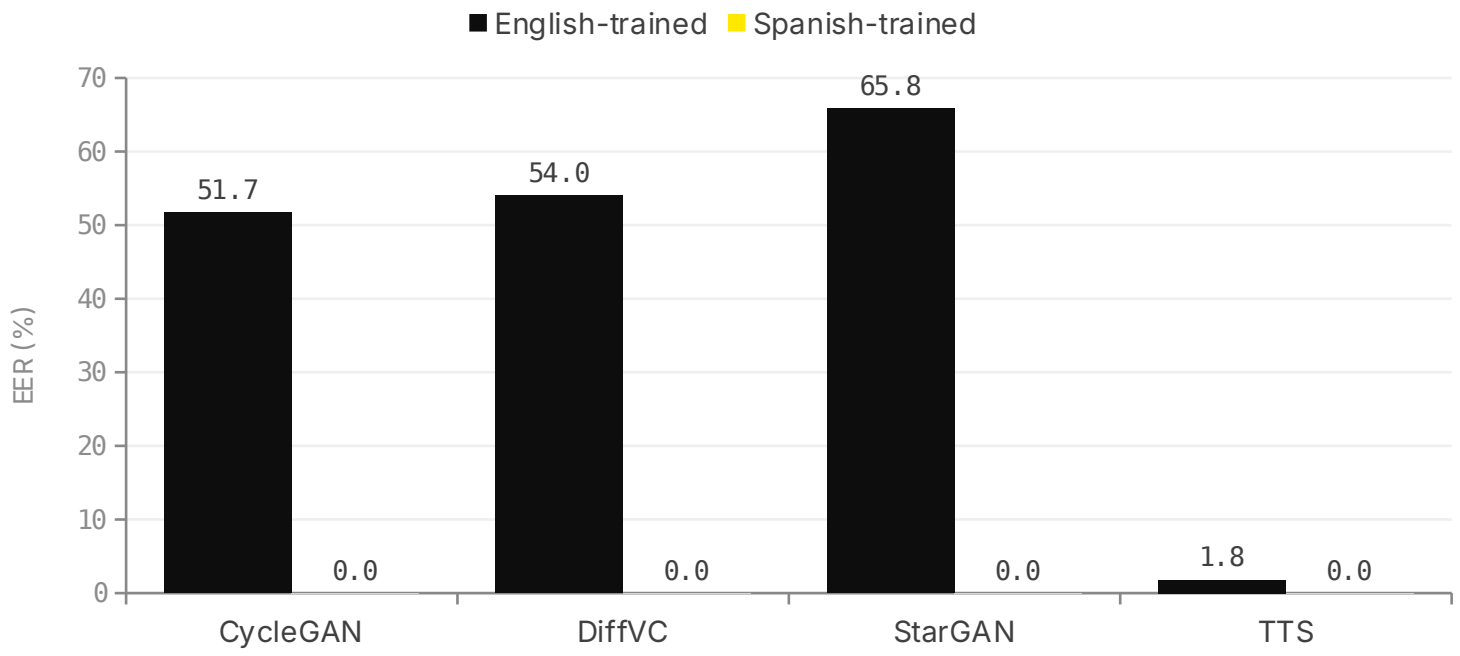


Table 6 · LCNN-LSTM-sum · P2S-Grad

50 % = a coin flip

English-trained models are at or below chance on converted speech — **and still catch Spanish TTS at 1.79 %.**

Vocoder artefacts survive the change of language. Conversion artefacts do not — they are entangled with the speaker.

1.79 %

EER on Spanish TTS attacks — from a model that never saw a word of Spanish.

The same model scores 43–66 % on voice conversion. That gap is the paper.

Accent barely moves it. Gender does.

Country	Overall	Female	Male
Chile	0.032	0.059	0.005
Peru	0.053	0.085	0.021
Argentina	0.140	0.177	0.103
Venezuela	0.168	0.065	**0.271**
Colombia	0.287	0.029	**0.544**

Table 10 · MFCC front-end · combined training

Two things to see

- **Accent spread**

  Below half a percentage point across all five countries.

- **Gender gap**

  Larger — and it reverses direction between countries.

The paper is titled 'accent-based'. Its answer about accent is: not much.

One of these is me. Which one?

A

[audio placeholder — embed .wav here]



B

[audio placeholder — embed .wav here]



Hands up for A · hands up for B

Then tell them the answer on the next slide — not this one.

Cross-lingual anti-spoofing failure is an attack-family problem, not a language problem.

Train on the language you will be attacked in — or at least on the conversion systems you will be attacked with.

THANK YOU

Questions — and what comes next

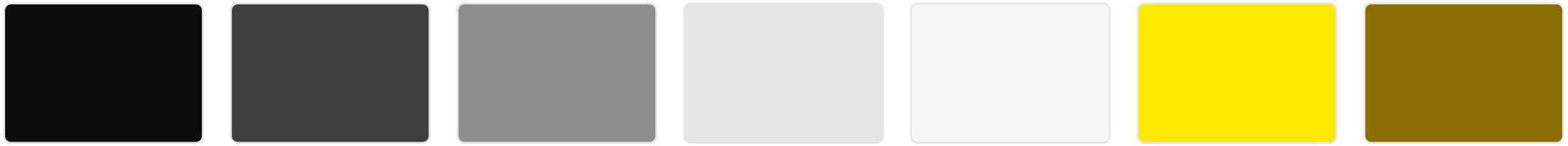
MARSA — a multi-accent Spanish corpus with six TTS systems, a partial-spoof tier, and an acoustic augmentation layer. Frame-level scoring, because utterance-level EER cannot express it.

QR
paper / code

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The system in one page



Ink
#0D0D0D
text, dark slides

Graphite
#3F3F3F
secondary text

Grey
#8C8C8C
captions, muted

Hairline
#E6E6E6
borders

Mist
#F7F7F5
card fill

Uniandes
#FFE800
accent only

Amber
#8A6D00
yellow as text

Type	Inter throughout. Title 40 / section 20 / body 13.5 / caption 10.5 / mono Consolas for numbers and sources.
Yellow	Accent only, never body text. On white, yellow that must be read becomes amber #8A6D00.
Motif	The highlighter bar. Three or four appearances in the whole deck, never two on one slide.
Grid	0.8" margins, 0.3" gutters, everything hangs from one left edge.
Rule	Every slide states its takeaway in one bold sentence. If you cannot write that sentence, cut the slide.